Price Variance: How Does Price Affect Sales?

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Abstract

This model evaluation aims to predict the change in sales based on a price variation of various products. However, these products were rather limited in price variation and did not strongly represent a relationship between price and quantity sold. The products provided, likely necessities, were not price sensitive. These factors may have been price insensitive but allowed for the discovery of non-linear relationships not visually presented in the available information. Gender and Tax were determined to be nonlinear relationships established in this dataset. Gender was determined in the random forest model due to its ability to analyze categorical data in non-linear relationships. An XGBoosted model tax was an important predictor as it is directly correlated to total price and quantity sold. Which is to be expected for this type of boosted model. Overall, linear models for this dataset were not statistically significant. Non-linear models where able to develop relationships due to their ability to handle complex relationships. Insightful business information was suggested as these nonlinear relationships where able to tell a story for business development. Going forward a stronger model can be developed with greater price variation of non-necessity items. While relationships were developed, it did not provide a linear relationship of sales based on quantity. To generate sales based on quantity, more business action should be taken to determine the effect of bottom line sales due to business implementation of the provided strategies.

Introduction

Price variation plays a vital role in the decision-making process for many reasons. The wrong decision could have severe implications resulting in various issues relating to finance, social, personal, or legal issues. Using this dataset I hope to explore, analyze, make predictions and review decisions based on this dataset. The goal is to explore and clean this dataset for interpretation, find patterns and trends, and evaluate the results. These efforts will create predictive models used for interpretation.

Data Collection

This data set was collected from Kaggle, an online platform where users can create, explore, and analyze data from various sources and posters. Kaggle is an online platform used for data science and machine learning practices to help others learn and develop their skills. The poster of this dataset created a fictional dataset for a company to target Key performance indicators and visualize information according to the present data. This dataset has multiple predictors all useful for predictive modeling.

Raw Data

The original data for this dataset comes with 11 predictors available for analysis. Of each of these predictors there are 1,000 data entries into this database. The first predictor in this dataset is the Sale ID. It has a range from 1 to 1,000. This verifies that this dataset does have 1,000 entries as the first entry starts at 1 and goes to 1,000. The second predictor is branch two branches are available A and B with most data entries falling in Branch A. This predictor is a predictor based on location or region of the supermarket the entry is in at time of data entry. City has the geographic location of 3 cities from the United States: Chicago, Los Angeles, and New

York. Customer type determines the type on shopper Member or Normal. Gender identifies Male or Female shoppers. Product name has the Variable of 5 different products available in this data set: Apple, Detergent, Notebook, Orange Juice, Shampoo. Unit Price is the price of the Item at the time of entry. Quantity is the number of items bought. Tax has the amount of tax charged per order. Total price is the cost with tax of the order. Finally, Reward points are the number of points earned for a transaction if applicable to the members.

The only other piece of information that was not included was product category this category appeared to have incorrectly labeled the items based on product category and seemed to make no logical sense. Therefore, this predictor needed to be removed from the dataset as to not create conflict for other analysis.

Data Cleaning

	sale_id	branch	city	customer_type	gender	product_name	unit_price	quantity	tax	total_price
1	1	A	New York	Member	Male	Shampoo	5.50	3	1.16	17.66
2	2	B	Los Angeles	Normal	Female	Notebook	2.75	10	1.93	29.43
3	3	A	New York	Member	Female	Apple	1.20	15	1.26	19.26
4	4	A	Chicago	Normal	Male	Detergent	7.80	5	2.73	41.73
5	5	B	Los Angeles	Member	Female	Orange Juice	3.50	7	1.72	26.22
6	6	A	Chicago	Normal	Male	Shampoo	11.24	9	7.08	108.24

Figure 1 Top of Dataset Information

After loading the dataset, the initial method to cleaning the data was to check the top of the dataset after removing the product category predictor. This allows for us to view the top of the dataset and briefly check for any errors that may have occurred between headers or data that would be apparent in the first few rows. Everything appears to be correct, and no errors shown.

```
'data.frame':    1000 obs. of  11 variables:
 $ sale_id      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ branch       : Factor w/ 2 levels "A","B": 1 2 1 1 2 1 1 2 1 2 ...
 $ city         : Factor w/ 3 levels "Chicago","Los Angeles",..: 3 2 3 1 2 1 1 2 1 2 ...
 $ customer_type: Factor w/ 2 levels "Member","Normal": 1 2 1 2 1 2 2 2 1 1 ...
 $ gender       : Factor w/ 2 levels "Female","Male": 2 1 1 2 1 2 2 1 1 2 ...
 $ product_name : Factor w/ 5 levels "Apple","Detergent",..: 5 3 1 2 4 5 5 5 1 5 ...
 $ unit_price   : num  5.5 2.75 1.2 7.8 3.5 ...
 $ quantity     : int  3 10 15 5 7 9 1 9 20 19 ...
 $ tax          : num  1.16 1.93 1.26 2.73 1.72 ...
 $ total_price  : num  17.7 29.4 19.3 41.7 26.2 ...
 $ reward_points: int  1 0 1 0 2 0 0 0 30 37 ...
```

Figure 2 data make-up

The next check for this dataset can be to check the data types to make sure each predictor is being read properly. Each predictor is normal and is met with no errors in this section. Following the check to make sure each predictor is appearing correct we can get a summary of the dataset which may tell a little more detail on the information being used.

```
    sale_id       branch           city     customer_type    gender          product_name
 Min.   :   1.0   A:674   Chicago    :330   Member:516    Female:472   Apple       :185
 1st Qu.: 250.8   B:326   Los Angeles:326   Normal:484    Male  :528   Detergent   :189
 Median : 500.5           New York   :344                               Notebook    :194
 Mean   : 500.5                                                        Orange Juice:208
 3rd Qu.: 750.2                                                        Shampoo     :224
 Max.   :1000.0
   unit_price        quantity          tax          total_price     reward_points
 Min.   : 1.020   Min.   : 1.00   Min.   : 0.080   Min.   :  1.21   Min.   : 0.000
 1st Qu.: 5.867   1st Qu.: 5.00   1st Qu.: 2.510   1st Qu.: 38.38   1st Qu.: 0.000
 Median :10.615   Median :10.00   Median : 5.870   Median : 89.70   Median : 0.000
 Mean   :10.836   Mean   :10.34   Mean   : 7.758   Mean   :118.58   Mean   : 6.057
 3rd Qu.:15.883   3rd Qu.:16.00   3rd Qu.:11.523   3rd Qu.:176.07   3rd Qu.:10.000
 Max.   :20.980   Max.   :20.00   Max.   :28.390   Max.   :433.99   Max.   :43.000
```

Figure 3 Summary Statistics

The summary chart shows various information that can be used to check for extreme values and get a general idea about the dataset. Each predictor has information that corresponds to Figure 2. This summary could have inaccurate statistics if the predictors are not accurately labeled. Upon initial inspection there are no extreme values for obvious outliers in this dataset currently. Each predictor in this table can give more detail in a brief descriptive table. Looking at each Predictor start to develop key patterns and indicators.

The First Predictor Sales ID shows a minimum, maximum, mean, median, first, and third quartile range. The minimum Value 1 and maximum value confirms there are 1 to 1,000 entries in this dataset. A mean and median of 500.5 would lead to believe that this data is consistent in the numbering with no errors. The first and third quartile values also follow similar patterns. Thia predictor is accurate in representing Sales ID in a chronological order.

The next predictor branch has two options in it, A and B. This predictor corresponds to the region or branch of the supermarket that the entry is in. This represents two locations which may allow for further analysis to how sales are impacted based on location in the supermarket as well. Between these two locations there is a total of 1,000 entries. Each predictor that responds to data entry values all equal 1,000 entries therefore this dataset currently has no errors in response predictors. This predictor also helps to identify that most items are in Branch A of the dataset. Further analysis can be conducted to determine if this predictor is significant.

Location also can play a large factor in the total price of each item for analysis. The city predictor has location for three major United States cities: Chicago, Los Angeles, and New York. Geographic information can be beneficial for making actionable insights based on a smaller local population when used for targeting price points for customers in the area. This chart reveals that the entry for different cities is relatively even.

Knowing the customers that are shopping in the store is a helpful tool for analytical insights as members are likely repat shoppers and can develop a pattern for shopping and spend analysis. Nonmembers or Normal as referred too in this dataset could have some repeat customers or be compiled of new clients. This information is also approximately even with slightly more customers being normal.

Knowing the products that are going to be analyzed is useful for many reasons. The five products available for this dataset is apple, detergent, notebook, orange juice, and shampoo. These various products can relate to other predictors allowing for further analysis. The products bought are not equal. The most purchased item is Shampoo and least purchased is apple.

Unit Price is important for this analysis as it is the price of each item. This predictor has a clear difference in price for each of these products. Without this variability in price a price analysis would not be compatible for this data set. Quantity refers to the amount of the product that was bought, and tax is the amount of sales tax applied to the sale. Total sales are the cost of the transaction with sales tax. Rewards points refers to the number of points members would earn for the transaction. This portion of data may be useful in different analysis questions, but not currently important to this model.

Finally, checking for missing values to ensure this dataset is explored thoroughly. There are no current missing values and based on the summary table that has been explore there are no errors due to incorrect information or errors. With no errors currently present it is vital to check the value of total price to verify that this predictor aligns with the variables it is inferred to use. Total price is comprised of unit price, quantity, and tax. This variable is vital to ensure accuracy in the dataset. Creating a calculated variable that multiplies the unit price, quantity, and tax shows that total price is accurate as both these predictors are equal. Since these two predictors are the same and free of errors then the calculated total can be dropped from the dataset as it is just created to verify accuracy.

Outlier detection

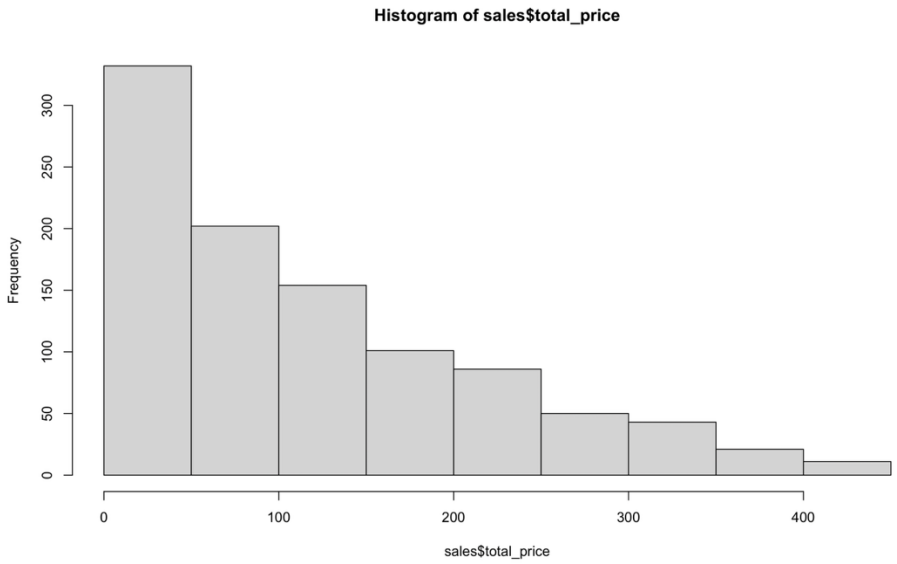

Figure 4 Histogram of Total sales based on Price

This dataset reported outliers outside of the 1.5 Interquartile Range. These Outliers however important are not errors or incorrect values. The outliers in this dataset is found in the tax, total price, and rewards points predictors. This suggests that large volume sales are not common and represent genuine business data. Figure 4 suggests this information to be accurate representing high volume of sales with lower volume of quantity purchased. This figure also shows a similar pattern in a histogram of the other predictors. This activity is business activity useful for helping to establish if price is a factor in these high quantity purchases. These values will be kept in this dataset due to accurate high quantity purchases being verified as true.

Log-Log Price Elasticity

```
Residuals:
    Min      1Q  Median      3Q     Max 
-2.1309 -0.4609  0.2460  0.6987  0.9517 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.13462    0.08453  25.253   <2e-16 ***
log_price   -0.03009    0.03684  -0.817    0.414    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8429 on 998 degrees of freedom
Multiple R-squared:  0.0006679,	Adjusted R-squared:  -0.0003335 
F-statistic: 0.667 on 1 and 998 DF,  p-value: 0.4143
```

Figure 5 Price Elasticity

Elasticity = % change in Quantity / % change in Price

The log-log price elasticity model represents the best model for price elasticity because it measures the percent change in quantity and percent change in price. This model is a ratio of the percentage of change. Log values help to reduce skew, reduce large values, and improve the model fit. This model shows there a -0.03% reduction in sales based on price. This represents a not statistically significant change in value. Therefore, price does not affect quantity sold in this dataset. With this insignificant price change controlling by product type, city, and customer type can help isolate the price effect.

Controlling for these additional predictors controls the price effect for item popularity, differences between cities, and differences in customer buying behavior. This Model predicts a 0.0277% change in the quantity sold after controlling based on product, city, and customer type. This verifies price insensitivity while increasing model predictability. The baseline product Apple sold less than the other products. Showing a difference in demand that is independent

from price. City shows that Los Angeles and New York sold less products than Chicago by approximately 12-13%. Customer type did show that Rewards customers purchased more often than non-members. This difference could create a meaningful change is sales by marketing their rewards system more. Overall, this model highlights the little influence on price and quantity sold. However, it does show that differences in demand are driven by product characteristics and location sold.

These two models may show little pricing significance but we can determine this is likely because these products are often considered necessities. These necessities can be influenced by location and differences in demand for these varying locations seem to be different.

Random Forests

A random forest model was chosen as the next step in predictive modeling due to the lack of linear relation between the earlier models. A random forest was chosen because of the ability it must measure non-linear relationships. Random forests help to capture nonlinear interactions and is better at handling the categorical value in this dataset. This model returned a variance of roughly 80% with a mean squared error of approximately 7.04 and RMSE of approximately 2.67. This means that our models prediction error is roughly 2.67 units off with approximately 80% of the variation in quantity. This created a much stronger predictive model with minimal overfitting and provides variable importance explaining gender to be one of the highest predictors.

Lasso Model

Utilizing a Lasso model for predictive analysis established that a linear relationship in this dataset is not a reliable method of prediction for this model. This Lasso model utilized the

10-fold cross validation method. The optimal penalty was lambda approximately 0.088. Utilizing the 1 SE rule the lasso model achieved a RMSE of approximately 3.58. The R squared for this method achieved a value of -21.70 indicating the model performance was far worse in prediction. Linear model selection is not a suitable method for this dataset.

XGBoost

Utilizing an XGBoost model will help predictive performance because it is a gradient boosted tree that can model the complex nonlinear interactions in this model. XGBoosting allows for optimized boosting. Including this model suited well for this dataset as this dataset had no linear relation as discovered previously. This model preformed the best overall preforming a near perfect R squared value of 0.9998. This extremely high value may suggest there is potential for overfitting in this model. It may be learning noise or other patterns rather than the generalized relationships.

	Feature	Gain	Cover	Frequency
	<char>	<num>	<num>	<num>
1:	tax	9.999465e-01	0.489842380	0.31493078
2:	unit_price	1.850730e-05	0.274684815	0.25280158
3:	quantity	1.068845e-05	0.072666306	0.08239947
4:	reward_points	8.296601e-06	0.096821125	0.05866842
5:	product_name.Notebook	3.963116e-06	0.004633745	0.02307185
6:	product_name.Orange Juice	2.304521e-06	0.006134942	0.02389585
7:	city.Chicago	1.572425e-06	0.009289551	0.03988134
8:	customer_type.Member	1.555069e-06	0.003756852	0.02999341
9:	product_name.Detergent	1.394427e-06	0.007375829	0.02488464
10:	branch.A	1.330331e-06	0.011309163	0.04746210
11:	gender.Female	1.318326e-06	0.007147506	0.03938695
12:	city.New York	1.311445e-06	0.005395925	0.02208306
13:	product_name.Shampoo	7.336922e-07	0.007294206	0.02109426
14:	product_name.Apple	5.121199e-07	0.003647654	0.01944628

Figure 6 Figures of Importance

This Variable importance chart reported tax as a important variable for reducing error. This is because tax should increase as quantity increases because these variables are directly related. This predictor also directly influenced how observations are impacted by splitting with

this predictor. Frequency shows that it was also most often used in these trees. This importance plot shows why the R-squared performed so well in this model. This model is verifying that tax directly correlates to changes in quantity because as tax increased quantity increased. XGBoost relies heavily on this variable when constructing trees because of relationship between these two variables. Conducting this model did not reflect error it reflects the transactional relationship of this dataset. Therefore, this prominent level of correlation didn't negatively impact the models function.

Model Comparison

These models helped to understand the values of this dataset. Conducting a log-log price variation allowed for testing for nonlinear models as linear model detection with price variance and lasso determined linear model development to be very ineffective and no linear relationships existed. This could be for several reasons as these products are likely products that are necessities and ability to model based on linear relationships were not strong predictors. If attempting to determine the effect of a price increase by quantity sold providing more data with a greater price variation between products would help to establish a better report on price variance. Providing non-essential items may help to create a better predictability. However, linear prediction may not have been established we were able to create nonlinear predictive models for this dataset that had reasonable statistics of prediction. Exploring different models between linear and non-linear helped to establish non-linear relationships between these charts. Each linear model essentially provided that these items are price insensitive and likely necessary. To test the strength of this price insensitivity would require significantly higher price variation. Product categories were rather broad having specific products within a category that have greater differences in changes may have made the price variation model much stronger. The random

forest model proved to be the first strong linear model preformed with 80% of the variation explained in this model. This model determined gender was a factor of importance while the XGBoost determined tax was the leading importance. With 99% explanation of variance. Tax directly correlated to the quantity sold therefore, this models findings show that tax was a leading factor of importance for building boosted models. With more data and variation in price further analysis on gender and sales differences may be determined. For statistical performance based models, the Boosted model preformed the best. This is to be expected due to the explanation of tax. Random Forest determined gender to be of important factors.

Choosing to still model the linear models helps to verify that these products are price independent. This was a verified due to the variance equation and linear model preformed. An expanded dataset with more information would allow for further analysis. With linear models not dependable valuable insights were able to be developed. These various prediction models developed strong analytical insights with great opportunities for advancement for increasing sales. While conducting these models it was discovered that the city played a large factor in driving sales. While also proving to increase sales with their rewards program. A strong business strategy can be developed with the information provided out of these models. Developing a business strategy can help to drive sales and report more sensitivity to price based on historic information. In the future focusing on the following strategy and providing an updated dataset will help to establish a more linear relationship.

Business Strategy for Improved Sales

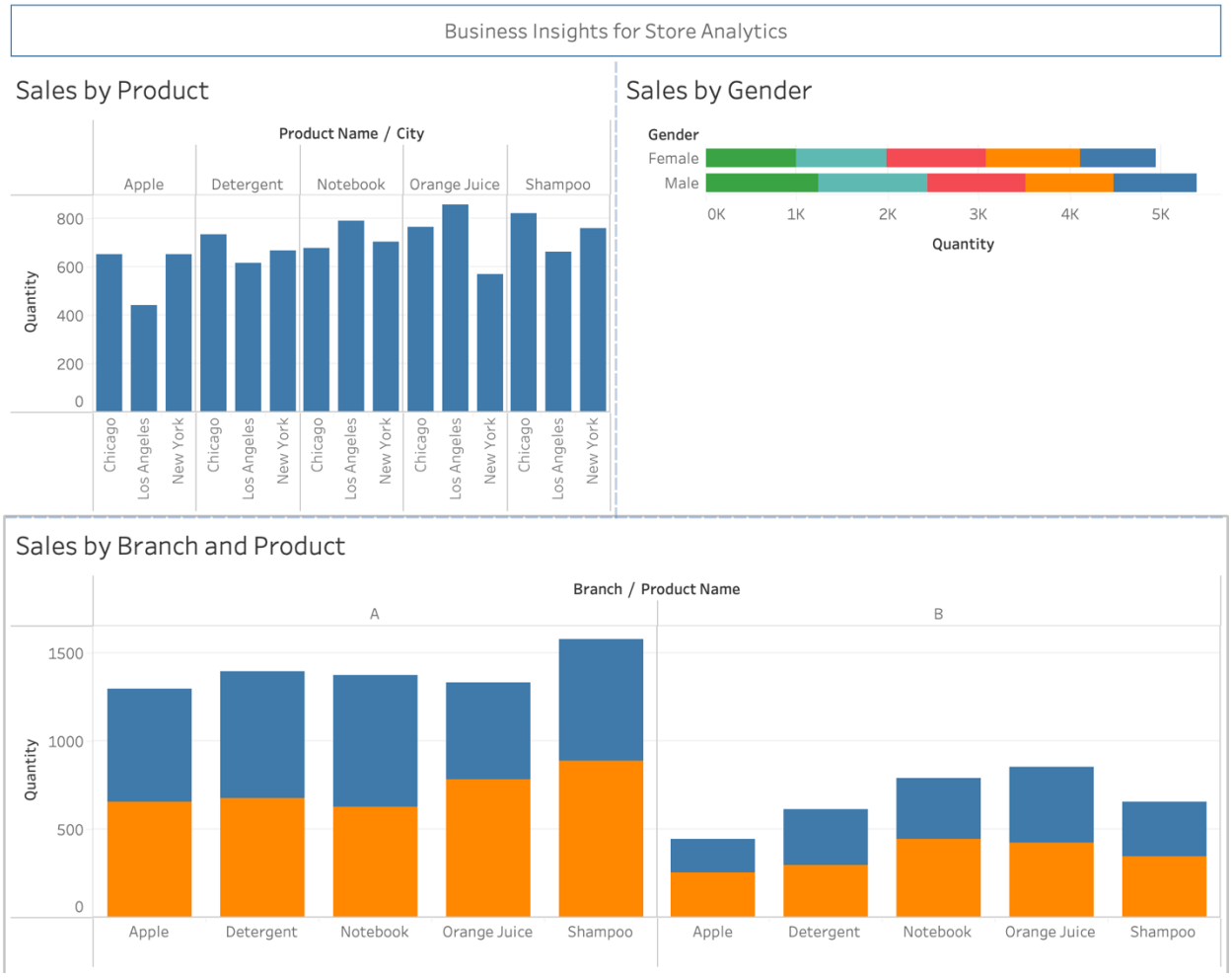

Figure 7 Data Visualization for Improved Business Insights

Utilizing Tableau to visualize some generic business information a relationship can be developed. Branch Sales are lowest in Branch B and highest in Branch A. City is also a strong influential visual factor as well, as determined earlier. Chicago is the highest selling store on average while Males tended to purchase more. The clear strategy to develop here is to market the rewards program more as members had a more frequent spend. Likely due to the points system in place creating a repetitive customer base. Limiting product availability in Branch B will also be ideal. Offering more selection in Branch A or variety may lead to increased sales overall. Sales are also 12-13% lower in Los Angeles and New York therefore stronger marketing needs to be done for these locations. Competition may be much higher and affecting the sales in these

regions. Increasing members enrolled into the membership is likely the best strategy as there was a direct correlation to members spend vs non-members spend. Overall, Targeting the membership program and increasing marketing expenses for the two lower selling cities is the best overall strategy for improved sales.

Future effective insights and recommendations

This dataset was rather limited in its predictability with price. Non-linear relationships were established for valuable information for modeling. In the future product expansion should be included to allow for more predictable results. Information provided included rather limited products and/or generic information. Specific products of each category i.e. Dove shampoo may help for a much greater price variation analysis. This dataset was great for generating key performance indicators of business administration of sales for this store. This analysis was useful for exploring non-linear relationships and developing key insights for how relationships are formed in a non-linear fashion. To generate better information based directly on sales, further information must be provided.

Reference page

Wambles, C. (2024). SuperMarket Sales. Kaggle.com.
https://www.kaggle.com/datasets/chadwambles/supermarket-sales/data?select=sales.csv